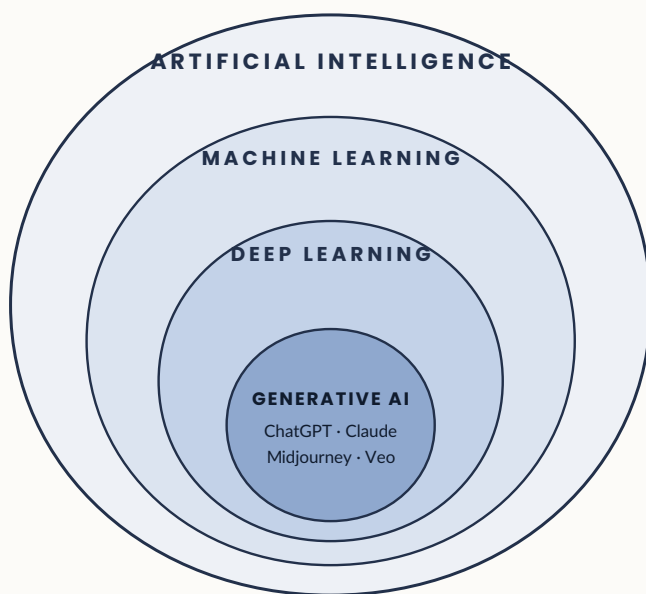


"AI", "Machine Learning", "Deep Learning" — who is inside whom?

The 60-second map, from someone who has taught this at university since 2021.



Artificial Intelligence

Any technique that makes machines do things that look intelligent. The umbrella — includes systems with zero learning.

e.g. a chess program following hand-written rules

Machine Learning

Systems that learn the rules from data instead of being programmed with them. Show examples, get behavior.

e.g. spam filter trained on millions of emails

Deep Learning

Machine learning with neural networks stacked in many layers — the approach behind almost every modern AI headline.

e.g. face recognition, speech-to-text, my autism research

Generative AI

Deep learning that produces new content — text, images, video, audio — instead of only recognizing patterns.

e.g. ChatGPT writes; Midjourney paints; Veo films

THREE SENTENCES PEOPLE WILL STOP SAYING TO YOU

"AI = robots"

Most AI is invisible software. Your email, your bank, your camera roll — all running ML. Robots are the rare, expensive exception.

"AI understands"

Generative AI predicts what plausibly comes next. Astonishingly useful — but confidence is not comprehension, which is why it can be confidently wrong.

"AI is new"

The field is from 1956. What is new is the data and the computing power that finally made deep learning practical — roughly since 2012.